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Github Project Repository: <https://github.com/I-go-Ope/Projects.git>

EDUCATION

Elmhurst University

Bachelor of Science in Engineering Physics
Bachelor of Science in Mathematics

Elmhurst, IL

May 2026

Relevant Coursework: Modern Physics, Mechatronics I & II, Analytical Mechanics, Thermodynamics, Electricity & Magnetism, Modern Optics, Data Science, Particle Astrophysics, Modern Particle Astrophysics, Modern Particle Astrophysics III, Quantum Mechanics, Differential Equations, Applied Analysis, Statistics for Scientists, Linear Algebra, Probability Theory

Relevant Coursework GPA: 3.76/4.0

PUBLICATIONS

M. Agostini *et al* [incl. **F. Fuchs**]. *Prototype acoustic positioning system for the Pacific Ocean Neutrino Experiment*. Journal of Instrumentation, 20(07), P07003 (2025).

<https://doi.org/10.1088/1748-0221/20/07/P07003>

O. Aghaei *et al* [incl. **F. Fuchs**]. *Long term study of sedimentation and biofouling at cascadia basin, the site of the pacific ocean neutrino experiment*.

European Physical Journal C, 86, 172 (2026).

<https://doi.org/10.1140/epjc/s10052-026-15319-6>

CONFERENCE PROCEEDINGS

D. Ghuman *et al* [incl. **F. Fuchs**]. *In Situ Calibration Systems for the Pacific Ocean Neutrino Experiment*.

Proceedings of Science (PoS), ICRC2025, 1042 (2025).

<https://doi.org/10.22323/1.501.1042>

V. Gousy-Leblanc *et al* [incl. **F. Fuchs**]. *Development and testing of the optical module for the Pacific Ocean Neutrino Experiment (P-ONE)*.

Proceedings of Science (PoS), ICRC2025, 1047 (2025).

<https://doi.org/10.22323/1.501.1047>

C. Lagunas Gualda *et al* [incl. **F. Fuchs**]. *First instrumented line of the Pacific Ocean Neutrino Experiment: status, development and outlook*.

Proceedings of Science (PoS), ICRC2025, 1087 (2025).

<https://doi.org/10.22323/1.501.1087>

C. Lagunas Gualda *et al* [incl. **F. Fuchs**]. *Status update of various simulation components for P-ONE*.

Proceedings of Science (PoS), ICRC2025, 1088 (2025).

<https://doi.org/10.22323/1.501.1088>

V. Alexis Parrish *et al* [incl. **F. Fuchs**]. *Pacific Ocean Neutrino Experiment first string trigger and data acquisition systems*.

Proceedings of Science (PoS), ICRC2025, 1141 (2025).

<https://doi.org/10.22323/1.501.1141>

B. Veenstra *et al* [incl. **F. Fuchs**]. *Development of monitoring and control strategies for biofouling and sedimentation for the Pacific Ocean Neutrino Experiment*.

Proceedings of Science (PoS), ICRC2025, 1202 (2025).

<https://doi.org/10.22323/1.501.1202>

INVITED TALKS

A. Magaña Ponce, **F. Fuchs**, R. Halliday. *Pacific Ocean Neutrino Experiment: Central Timing*.

Presented to the Elmhurst University Board of Trustees, Elmhurst, IL, February 2025

ORAL PRESENTATIONS

F. Fuchs, *Composition of Ultra High Energy Cosmic Rays (UHECR)*.

Senior Paper Presentation, Elmhurst University, Elmhurst, IL, May 2026

F. Fuchs. *Absorption and Dispersion*.

Research Seminar Series, Elmhurst University, Elmhurst, IL, December 2025

A. Magaña Ponce, **F. Fuchs**, R. Halliday. *Pacific Ocean Neutrino Experiment: Central Timing*.

Research & Performance Showcase, Elmhurst University, Elmhurst, IL, May 2025

F. Fuchs. *Electrostatic Analog II*.

Research Seminar Series, Elmhurst University, Elmhurst, IL, May 2025

F. Fuchs. *Electrostatic Analog I*.

Research Seminar Series, Elmhurst University, Elmhurst, IL, December 2024

F. Fuchs. *Coupled Oscillations (w/ Educational Aid)*.

Welcome to STEM Poster Session, Elmhurst University, Elmhurst, IL, September 2024

A. Magaña Ponce, **F. Fuchs**, R. Halliday. *Pacific Ocean Neutrino Experiment: Central Timing*.

CASE Showcase, Elmhurst University, Elmhurst, IL, August 2024

F. Fuchs. *Coupled Oscillations (w/ Education Aid)*.

Research Seminar Series, Elmhurst University, Elmhurst, IL, December 2023

POSTER PRESENTATIONS

F. Fuchs, T. Imanlihen, R. Halliday. *Cosmic Ray Shower Development By Element at 10^{18} eV*.

Welcome to STEM Poster Session, Elmhurst University, Elmhurst, IL, September 2025

A. Magaña Ponce, **F. Fuchs**, R. Halliday. *It's About Time... and Neutrinos: Synchronizing the Pacific with P-ONE*.

Summer Research Showcase, Elmhurst University, Elmhurst, IL, September 2024

F. Fuchs, A. Magaña Ponce, R. Halliday. *Composition of Ultra High Energy Cosmic Rays*.

Research & Performance Showcase, Elmhurst University, Elmhurst, IL, May 2026

ACADEMIC RESEARCH

Development of the P-ONE Neutrino Detector, Elmhurst University
Undergraduate Research (Fellow) under Dr. Robert Halliday

Elmhurst, IL
May 2024 - December 2025

- Developed and implemented a custom central timing board for the Pacific Ocean Neutrino Experiment (P-ONE), enabling sub-microsecond synchronization across distributed optical detector modules
- Utilized serial communication interfaces to calibrate synchronization offsets across multiple timing channels, ensuring consistent phase alignment and stable timing performance throughout the detector system
- Designed and executed tests of data acquisition and signal-propagation pathways, achieving a measured timing precision of ~800 femtoseconds, significantly surpassing the expected ~1 nanosecond system requirement
- Contributed to system-level integration, validation, and deployment for the first P-ONE-1 detector string in the Victoria Basin in Canada

Ultra-High Energy Cosmic Ray Shower Simulations, Elmhurst University

Elmhurst, IL

Undergraduate Research under Dr. Robert Halliday

February 2025 - Present

- Ran Corsika simulations across several different devices with focus on the relationship between the energy of incoming particles and the atomic species interacting with atmospheric particles, and analyzing the maximum number of particles to come from this collision
- Ran a series of different test to see if atomic species can be determined based on shower depth
- Created documentation allowing for future students to engage with this project with little to no confusion

LEADERSHIP & SERVICE

Math & Physics Tutor, Elmhurst University

February 2024 - May 2024

- Tutored students taking introductory and upper level math and physics courses, strengthening critical thinking and problem solving skills
- Worked with the students to individualize each session, formatting the information to be more digestible for each individual

Resident Advisor, Elmhurst University

August 2024 - May 2026

- Supported a residential community of diverse individuals serving as a resource for varying needs
- Responded to crises during on call shifts in collaboration with many campus departments
- Fostered community through event coordination and resource distribution

GRANTS, HONORS, & AWARDS

HIP Grant, Elmhurst University

May 2025 - August 2025

Supported contributions to research in (ultra) high energy cosmic ray shower simulations

HIP Grant, Elmhurst University

May 2024 - August 2024

Supported contributions to the P-ONE project

CASE Fellowship, Elmhurst University

May 2024 - August 2024

Funded research on the development of the P-ONE neutrino detector

Dean's List, Elmhurst University

February 2024 - May 2026

Recognized for academic excellence each semester since the spring of 2024

The Earl Swallow Memorial Chairperson's Book Award, Elmhurst University

May 2026

Recognized for excellence in research, academics, and leadership in the Department of Physics.

TECHNICAL SKILLS

Programming

Python, Arduino IDE, C++, Mathematica

Operating Systems

Windows, Linux (Ubuntu)

Simulation & Design Software
Corsika, RhinoCAD, KiCad

Serial Communications, Oscilloscopes, Function Generator

Laboratory & Instrumentation

Programs
Tableau, Tableau Prep, Logger Pro, TeXwork

PROJECTS

Portable Platform for P-ONE Timing System

- Designed and fabricated custom mounting stands for the P-ONE timing system to ensure stable and repeatable sensor positioning
- Developed a portable, integrated platform to secure all components, reducing setup variability and improving system organization
- Engineered a cable management solution using integrated Velcro routing to separate and direct power and data lines, minimizing clutter and signal interference
- Optimized system layout for rapid deployment, teardown, and transport between locations without requiring reconfiguration

Coupled Oscillations Educational Aid (With Linear Actuator)

- Created a physical display, using springs, dynamic carts and track, to develop physical intuition for the concept of coupled oscillations
- Designed, created, and incorporated a linear actuator to the system, allowing for students to study how driving forces affect the system
- Created code using Mathematica to show how the Aid should act under different parameters and boundaries, connecting the mathematical interpretation to physical interpretation

Cloud Chamber

- Designed and built a diffusion cloud chamber using plexiglass and metal plates
- Created a supersaturated alcohol vapor environment using dry ice cooling
- Visualized alpha and beta radiation through condensation trail formation
- Analyzed particle tracks to distinguish between radiation types

Small-Scale Rocket Design Project

- Designed and constructed a water-propelled bottle rocket using 2-liter bottles, floral foam, string, and custom 3D-printed fins
- Modeled and fabricated aerodynamic fins using RhinoCAD to improve flight stability and trajectory control
- Engineered a protective housing system for a Bluetooth motion tracker, ensuring data collection without compromising structural integrity
- Developed a secure mechanical connection (string-based reinforcement) to prevent structural separation under launch stresses

Low-Cost Interferometer Kit

- Designed and built an accessible interferometer for undergraduate optics experiments, enabling visuals for interference patterns